

Education Student Performance Analysis - Project Report

Project Overview

This project analyzes student academic performance using Python data analysis tools.

The dataset contains student information such as gender, course, study hours per week, attendance percentage, previous score, final score, and result.

The goal of the project is to identify patterns affecting student performance and generate insights that help educators understand how study habits and attendance impact final scores.

Setup Instructions

1. Install Python 3.x
2. Install required libraries:

```
pip install pandas matplotlib
```
3. Download the dataset: `education_student_performance.csv`
4. Run the notebook or Python script to perform analysis.
5. Execute visualization cells to generate charts.

Code Structure

education_project/

|

|— education_student_performance.csv

|— Education_Data_Analysis.ipynb

|— analysis.py

|— report.docx

Main Components:

- Data Loading
- Data Exploration
- Data Cleaning
- Visualization
- Insight Generation

Methodology

1. Load dataset using pandas.
2. Inspect dataset using head(), info(), describe().
3. Validate missing values using isnull().sum().
4. Perform statistical analysis on study hours, attendance, and scores.
5. Generate charts to visualize relationships between variables.
6. Extract meaningful insights from the analysis.

Technical Implementation

Libraries Used:

- Pandas: Data manipulation and analysis
- Matplotlib: Data visualization

Key Steps:

- DataFrame creation from CSV
- Statistical summary generation
- Feature exploration
- Graphical visualization of relationships between variables

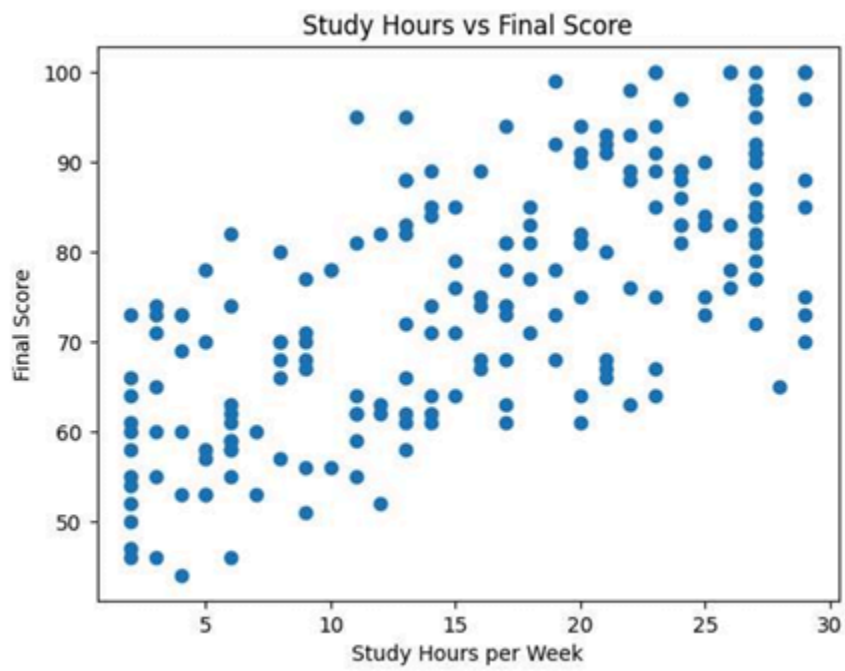
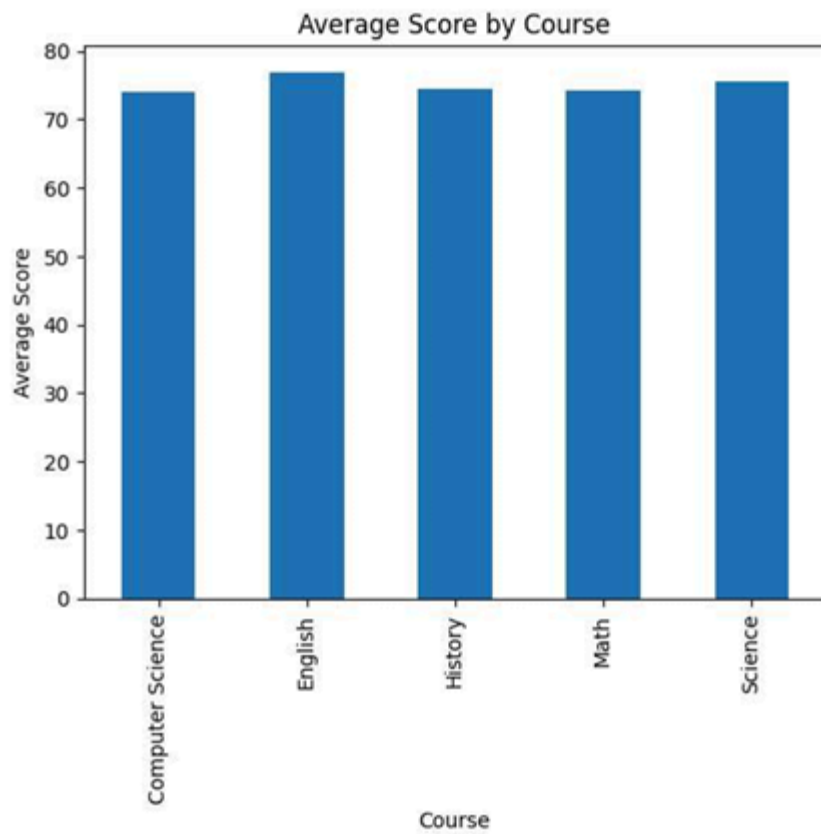
Business Insights

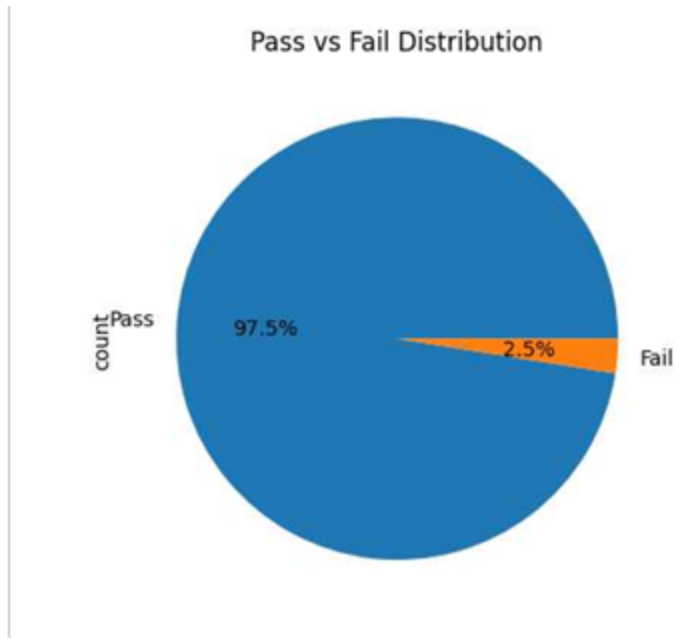
Key insights that can be derived:

- Students with higher study hours tend to achieve higher final scores.
- Attendance percentage strongly correlates with academic performance.
- Previous academic performance is a predictor of final results.
- Course selection may influence score distribution depending on difficulty level.

Visual Documentation

Visualizations included in the project:





These visuals help interpret patterns in the dataset.

Testing Evidence & Test Cases

Test Case 1:

Input: Dataset loaded

Expected Result: DataFrame successfully created

Test Case 2:

Input: `df.info()`

Expected Result: Correct column structure displayed

Test Case 3:

Input: `df.isnull().sum()`

Expected Result: No missing values detected

Test Case 4:

Input: Visualization generation

Expected Result: Graphs generated without errors

Conclusion

The Education Student Performance Analysis project demonstrates how data analytics can be used to evaluate student performance and identify factors affecting academic success.

Using Python, pandas, and matplotlib, we were able to analyze the dataset, generate visual insights, and draw conclusions that can support data-driven decision-making in educational environments.